Mohammed Eid Abdelmeguid

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Portfolio | GitHub | LinkedIn | Hugging Face

Summary

AI/ML engineer with Python backend experience and internship contributions to Sprints' AI digital twin. Builds document-retrieval systems, LLM applications and Arabic NLP pipelines. Co-author of an ArabicNLP 2026 shared-task paper; completed the Digilians AI & Data Science diploma in September 2026.

Experience

AI Intern | Sprints 4 Oct – 15 Nov 2026

- Worked on SprintFlow's Digital Twin proxy mode and multi-format data ingestion for RAG using FastAPI, LangGraph, PostgreSQL and Qdrant.
- Built file uploads, parsing, chunking, embedding/indexing and cited retrieval; contributed owner-controlled proxy replies and the knowledge used to ground them.

Freelance Backend Developer | StockGo (Qatar) Nov 2024 – Apr 2025

- Built and deployed a Django REST API and PostgreSQL-backed analytics dashboard for a live inventory-tracking product.

Full Stack Developer Intern | NfrtiX Mar – Apr 2025

- Built an internal Django/JavaScript application to support operational workflows.

Backend Developer Intern | Venture Appital Oct 2024 – Feb 2025

- Developed modular Flask REST microservices and unit tests.

Selected Projects

SprintFlow – Digital Twin | Sprints Internship Project

- AI workspace with proxy replies and a file-to-answer RAG pipeline combining semantic search with PostgreSQL ownership checks. My contributions focused on proxy mode and data ingestion; source repository is private.

AI Compass | Multi-Agent RAG & Recommendations May 2026 – Present

- Led a team of five on an AI news platform with seven ingestion sources, five enrichment agents, semantic search and a streaming RAG assistant.
- Built pgvector candidate retrieval, weighted ranking and MMR diversification with an offline NDCG@10/MAP evaluation harness; integrated Django, Celery/Redis and Groq-served LLMs.
- Deployed on Vercel and AWS EC2 with PostgreSQL; backend subsequently paused to control hosting costs.

Palestinian Cultural Knowledge Platform | Arabic NLP & RAG Jun – Aug 2026

- Owned ingestion, Arabic NLP, retrieval and knowledge-graph work on a two-person project. Published a bilingual corpus of 882 documents (about 890K words) on Hugging Face with gated file access.
- Built CAMeL Tools NER (F1 0.47), a retriever (Recall@5 0.93 across 1,282 passages), and a graph of 13,063 canonicalized entities; evaluation figures are project-specific.

Selected Projects (continued)

AynVQA-MSA | Arabic Spoken Visual Question Answering — Jul – Oct 2026

- Implemented a modular Whisper ASR, schema-constrained parsing, conditional repair and joint vision-language answer-selection pipeline for the ImageEval 2026 MSA shared task.
- Published frozen-system results: **65.6% accuracy on 1,000 blind-test examples** (57.4% balanced accuracy; 50.4% macro F1), versus 79.0% on 500 development examples.
- Co-authored the shared-task paper analyzing the synthetic-to-human recording shift and a caching reproducibility issue; most development improvement came from the answer-selection model change.

Fitness Activity Tracker | Sensor-Based Machine Learning — Mar – Apr 2026

- Preprocessed wearable sensor data, engineered PCA/FFT features, compared five classifiers across five feature sets, and implemented independent repetition counting on a three-person team.

Publication

Digilians at ImageEval 2026 Shared Tasks: Modular Speech Parsing and Joint Multimodal Reasoning for Arabic Spoken VQA

Ahmed Eid Hassan, Nour Eldeen Hossam, Ahmed Hossam Rashad, Abdelrhman Mahmoud Fawzy, **Mohammed Eid Abdelmeguid**, Mohamed Hamdy Mohamed. *ArabicNLP 2026: Shared Tasks*, pp. 766–770, October 2026. DOI: 10.18653/2026.arabicnlp-sharedtasks.96.

Technical Skills

AI Agents & RAG: LangGraph, embeddings, pgvector, Qdrant, Ollama, Groq API, structured outputs, document retrieval
Machine Learning: Scikit-learn, XGBoost, feature engineering, model evaluation, grid search; TensorFlow/Keras; familiarity with PyTorch
NLP & Speech: CAMeL Tools, NER, BERTopic, NetworkX/Wikidata, Whisper, Qwen vision-language models
Backend & Data: Python, SQL, FastAPI, Django, Flask, PostgreSQL, Celery, Redis, SQLAlchemy/SQLModel, Pandas, NumPy
Tools: Docker, Git, GitHub Actions, Linux, Jupyter, Streamlit

Education

Alexandria University – Faculty of Engineering — Sep 2019 – Jun 2024
BS in Communication and Electronics Engineering

- **ADAS graduation project**: A+, ranked 5th of about 60 in the **Valeo competition**. Led software architecture and computer vision; mentored by Valeo and sponsored by ITIDA's Egypt Makes Electronics program.

Selected Certifications & Training

AI & Data Science Diploma – Digilians, MCIT scholarship — Sep 2026
AWS Machine Learning Engineer – Associate — Sep 2026
Huawei HCIA-AI — Aug 2026
IBM AI Engineering Professional Certificate — Jun 2026
Machine Learning Specialization – Stanford / DeepLearning.AI — May 2026
Microsoft Data Engineer – DEPI (159 hours) — Oct 2024